NM

NegoMaster

FINANCIAL NEGOTIATION PLATFORM

OFFICIAL SYSTEMS & VISUAL MANUAL

Multi-Bank Platform Architecture & Panel Operations Guide

An executive technical report, visual UI walkthrough, and step-by-step panel handling manual illustrating the multi-tenant architecture, live user interfaces, and delegated banking administration implemented in NegoMaster.

CLIENT PROPOSAL REFERENCE	SYSTEM DELIVERY STATUS	PUBLICATION DATE
NegoMaster Multi-Bank Architecture	Tier 1 Multi-Tenancy Live & Verified	October 2026

1. Executive Summary & Architectural Alignment 01

Following the target architecture proposed in the client's presentation, this document outlines how the vision of "*One Platform, Distinct Banking Environments*" is fully realized in the live platform. Central administration maintains governance over shared products, licenses, and global templates, while granting each banking tenant dedicated administration over its employees, managers, customized policies, skill assessment exams, and private audit trails.

> **Core Architectural Principle: Logical Multi-Tenancy (Tier 1)**
> All bank tenants share a single high-performance codebase and database infrastructure while maintaining strict, server-enforced data boundaries. Every database query, user session, policy, quiz, and log entry is strictly scoped by `business_id`. Zero cross-tenant data leakage is permitted.

TARGET MULTI-BANK SYSTEM ARCHITECTURE



Role-Based Access Control (RBAC) Hierarchy

Role	Portal & Guard	Permissions & Scope	Data Visibility
NegoMaster Super Admin	`/admin` (web guard)	Full platform authority. Tenant creation, subscription plan assignment, quotas, global templates, tenant impersonation.	Global (All Tenants)
Bank Administrator	`/business` (business.auth + business.admin_only)	Full tenant administration. Manages Bank Managers, employee directory, user approvals, policies, quizzes, and privacy policy.	Bank-Confidential (Own Tenant Only)
Bank Manager	`/business` (business.auth)	Operational evaluation role. Reviews employee exam submissions, grades custom responses, and views assessment analytics.	Bank Team Results (Read & Score)
Bank Employee / End User	`/api` & Mobile App	Learner role. Registers under bank code, reviews assigned policies, completes personalized profiling & skill exams.	Individual Results Only

2. Implementation Status: PPT Vision vs. Delivered System 02

This section provides a direct mapping against the client's proposal matrix (Slide 4 in the PPT), confirming that all 12 modules requested by the client are fully operational:

Module / Requirement	Status	NegoMaster Central (Super Admin)	Bank Administration (Tenant)
1. Dashboard	LIVE & OPERATIONAL	Platform-wide KPIs: active businesses, total users, global subscriptions, system health.	Bank KPIs: active employees, total exams completed, pending user approvals, active plan.
2. User Management	LIVE & OPERATIONAL	Global user directory, account status toggling, cross-tenant governance.	Employee CRUD, bulk Excel import, user registration approval pipeline (Pending/Approved/Rejected).
3. Bank Managers	LIVE & OPERATIONAL	Tenant oversight & security policy enforcement.	Dedicated manager creation with department assignment for exam grading delegation.
4. Business Provisioning	LIVE & OPERATIONAL	Create/edit banks, assign plan, quota, generate setup tokens, toggle Active/Suspended.	Tenant profile management, custom logo/contact details, privacy policy configuration.
5. Login As Business	LIVE & OPERATIONAL	One-click impersonation to access bank portal for support with seamless switch-back.	Notification bar showing active impersonation session with instant return button.
6. Profiling Environment	LIVE & OPERATIONAL	Standard global framework, section ordering, default question bank.	Custom sections, bank-specific questions, ordering, Excel import/export.
7. Business Policies	LIVE & OPERATIONAL	Global baseline policy templates and default case study questions.	Bank custom policies, welcome message, bilingual (EN/FR) policy sections, custom case studies.
8. Skill Assessment	LIVE & OPERATIONAL	Master assessment engine, standard exam templates, global question bank.	Bank-specific exam templates, custom question banks, pass mark configuration.
9. Results & Scoring	LIVE & OPERATIONAL	Global performance benchmarks and cross-bank analytics.	Detailed employee submission review with manual/auto scoring interface for admins and managers.
10. Contact Us Routing	LIVE & OPERATIONAL	Central support inbox for general inquiries & fallback routing.	Bank-specific contact inbox routing directly to the bank's configured email and portal.
11. Activity Logs & Audit	LIVE & OPERATIONAL	Global audit log (`AdminActivityLog`) tracking all central actions.	Bank-scoped audit log (`BusinessActivityLog`) recording bank events.
12. Settings & Privacy	LIVE & OPERATIONAL	Global platform settings, Stripe plans, default terms & conditions.	Bank-specific privacy policy with bilingual editor and locale switcher.

Summary of System Readiness

100% of the functional modules specified in the client's administrative matrix are implemented and operational. Both the Central Admin and Bank Admin portals provide clear, segregated workflows ensuring institutional control, data privacy, and operational ease.

2B. Architecture & Infrastructure Tier Analysis

02B

In Slide 9 and Slide 10 of the client proposal, three potential infrastructure tiers were presented: *Tier 1 (Logical Isolation)*, *Tier 2 (Dedicated Database per bank)*, and *Tier 3 (Dedicated deployment cluster per bank)*. This section explains the engineering rationale and why Tier 1 was implemented.

> **Architecture Decision: Why Tier 1 Logical Multi-Tenancy is Implemented**
>
> **Tier 1 (Logical Isolation)** is the industry-standard architecture adopted by leading enterprise platforms (e.g. Salesforce, Workday, Stripe). It achieves 100% data confidentiality while avoiding the massive operational overhead, reliability risks, and costs of running hundreds of isolated database instances.

Infrastructure Tier Comparison Matrix

Dimension	Tier 1: Logical Isolation (Delivered)	Tier 2: Dedicated Database (Roadmap)	Tier 3: Dedicated Cluster (Enterprise)
Data Isolation	Server-enforced via `business_id` across all tables and queries. Zero data leakage.	Separate physical database per bank; separate connection pools.	Completely isolated VPS/Kubernetes cluster per bank.
Hosting & Cost	**Optimal**. Low footprint, efficient resource sharing, easily scalable.	**High**. 300-400% cost increase due to multiple database engines.	**Extreme**. 800%+ cost increase; requires dedicated DevOps team.
Database Migrations	**Instantaneous**. Single atomic migration updates schema across all tenants.	**Complex**. Must iterate through dozens of databases; failure risk during upgrades.	**Manual**. Multi-cluster CI/CD pipeline required for every release.
Platform Maintenance	Central updates, unified bug fixing, zero downtime deployments.	Requires database sharding middleware and dynamic connection switching.	Requires 24/7 infrastructure monitoring per cluster.
Suitability	**Immediate Production Ready (Recommended)**	Future tier for high-compliance enterprise banks requiring dedicated DB.	Contractually mandated government / sovereign banking deployments.

Security Guarantees of Tier 1 Implementation

1. Server-Side Scoping, Never UI Filtering

Every Eloquent query in controllers and repositories automatically appends `where('business_id', $businessId)`. Tampering with client parameters or URLs cannot access another bank's records.

2. Separate Middleware Authentication

The business portal is protected by `business.auth`, which uses a dedicated session guard isolated from standard web users and central administrators.

3. Dual Audit Logging

Platform events are recorded in `AdminActivityLog`. Bank events are stored in `BusinessActivityLog` with strict tenant scoping. Banks cannot inspect each other's audit trails.

4. Future Migration Readiness

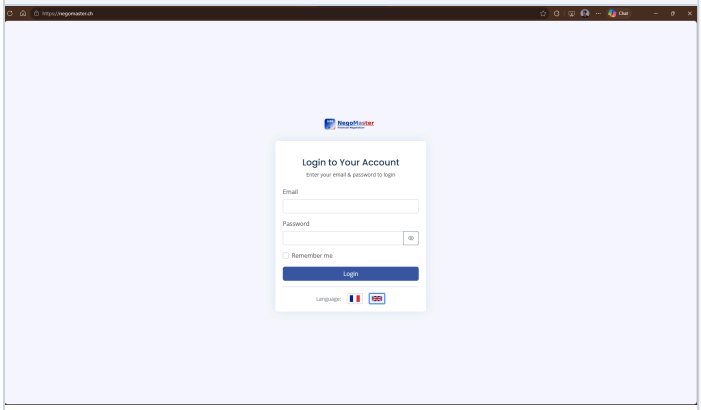
Because database interactions are decoupled through Repository classes, routing a specific enterprise client to a separate physical database in future phases requires zero frontend code rewriting.

3A. Super Admin: Login, Master Dashboard & User Governance

03A

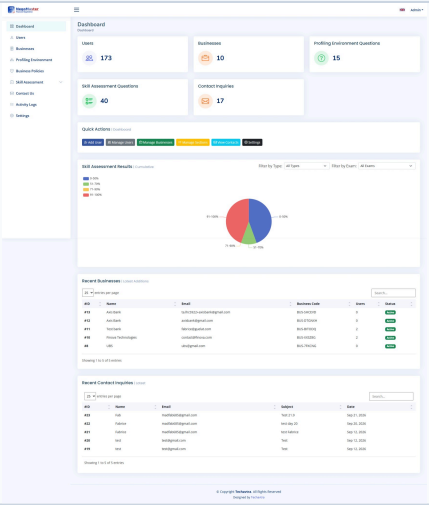
Super Admin Login Portal

/



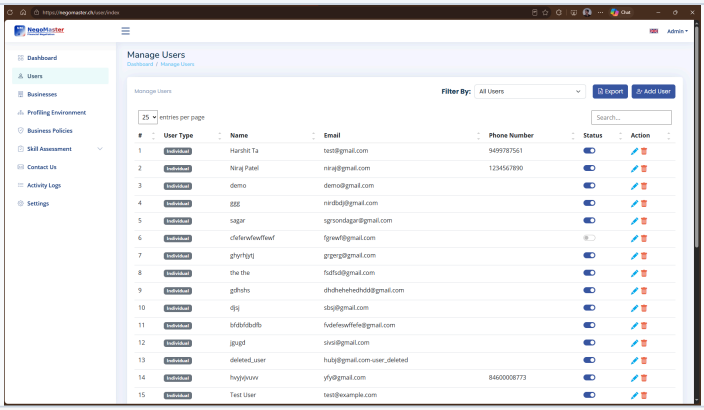
Super Admin Master Dashboard

/dashboard



Global User Management Directory

/user/index



OPERATOR INSTRUCTIONS: SUPER ADMIN DASHBOARD & USER GOVERNANCE

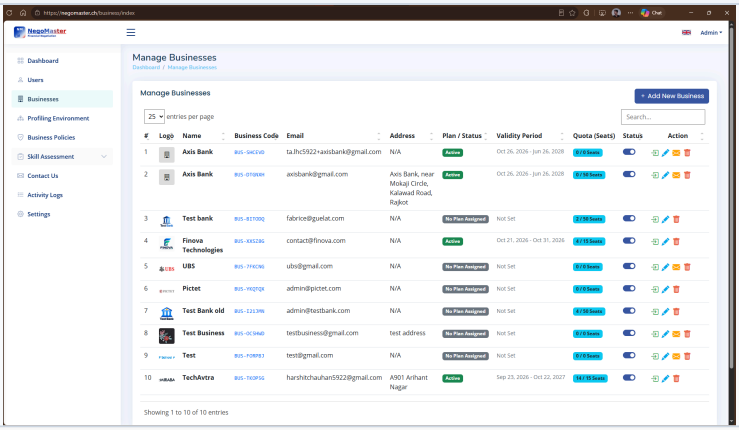
1. **Authentication**: Access the central login at root domain (/). Enter Super Admin credentials with remember-me support.
2. **Master Dashboard Oversight**: The dashboard provides real-time platform metrics: Total Registered Users, Active Business Tenants, Platform Subscriptions, and Recent System Activity logs.
3. **User Governance**: From **Users → Manage Users**, the administrator can inspect all user accounts across tenants, search by email/name, toggle user status (Active/Blocked), and review profile details.

3B. Super Admin: Bank Tenant Provisioning & "Login As" Feature

03B

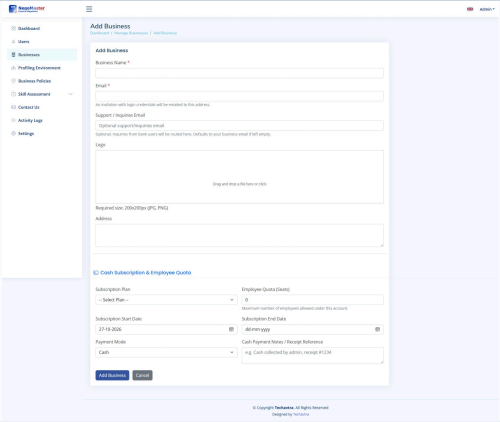
Business / Bank Tenants Directory

`/business/index`



New Business / Bank Provisioning Form Modal

`/business/create`



OPERATOR INSTRUCTIONS: BANK TENANT PROVISIONING & "LOGIN AS" SUPPORT

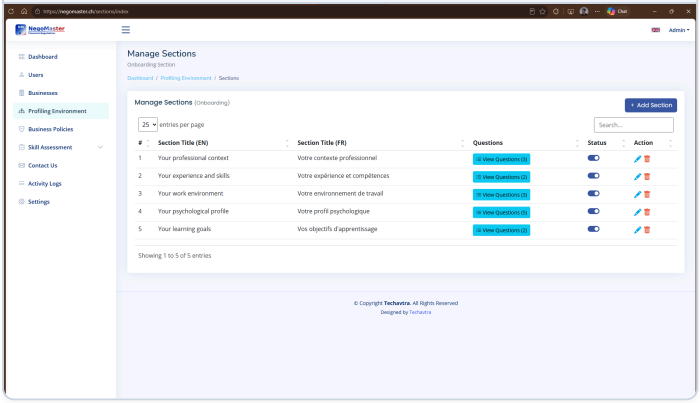
1. **Provision New Bank**: Click **"+ Create Business"**. Enter the Bank Name, Official Email, Subdomain/Slug, and initial quota settings. The system automatically creates a unique tenant ID and generates a password setup token sent via email.
2. **Configure Plans & Quotas**: Assign commercial subscription plans (Free, Pro, Enterprise) and define the monthly case analysis quota limit per tenant.
3. **Status Toggle**: Click **"Change Status"** to temporarily suspend a tenant. Suspended tenants immediately block access for their administrators and employees.
4. **One-Click "Login As Business" Support**: Click the blue **"Login As"** button next to any bank. You are instantly authenticated into that bank's live portal as their Bank Admin without needing credentials. A persistent banner allows switching back to Central Admin in one click.

3C. Super Admin: Profiling Environment & Baseline Policies

03C

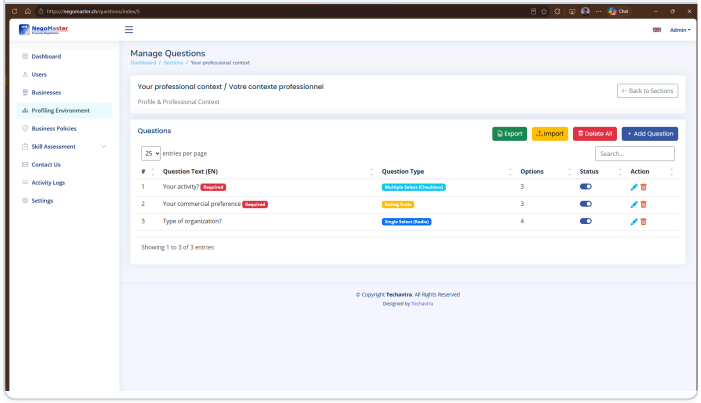
Profiling Environment Sections

`/sections/index`



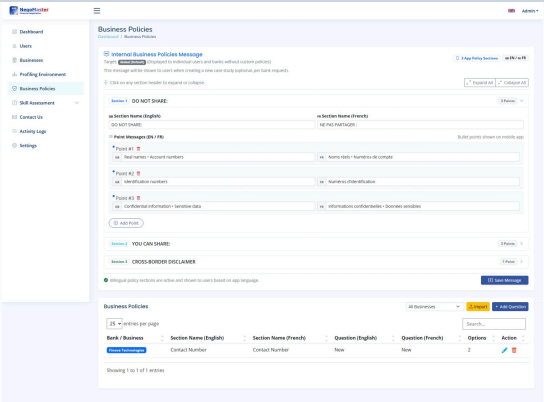
Profiling Question Management

`/questions/index/{id}`



Global Case Study & Policy Management

`/case-study-questions`



OPERATOR INSTRUCTIONS: MANAGING GLOBAL PROFILING & MASTER POLICIES

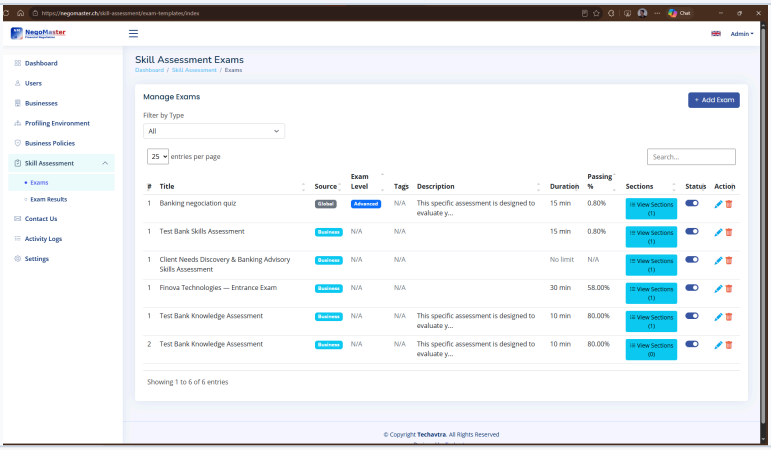
1. **Profiling Sections**: Organize behavioral and cognitive profiling sections. Drag-and-drop handles enable custom question ordering.
2. **Question Bank Authoring**: Add questions with custom response weights. Import or export question pools via Excel spreadsheets.
3. **Baseline Policies**: Define standard banking negotiation case studies. Banks automatically inherit these master guidelines unless they author local custom overrides.

3D. Super Admin & Bank: Skill Assessment Exam Engine

03D

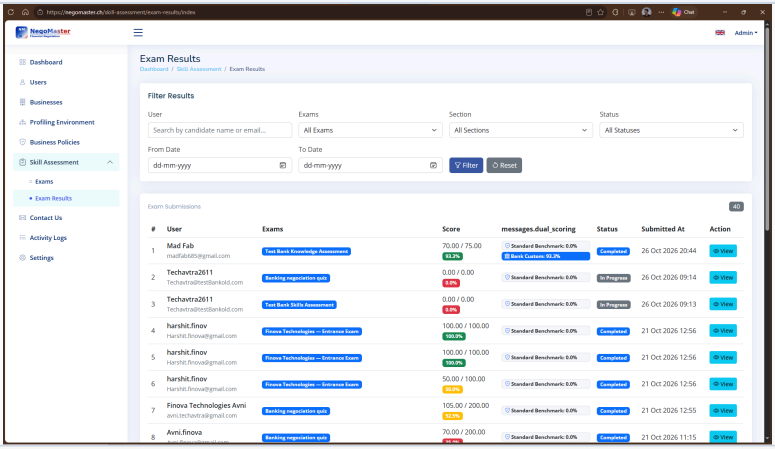
Super Admin Master Exam Templates

`/skill-assessment/exam-templates/index`



Bank Tenant Exam Management Portal

`/business/skill-assessment/exams`



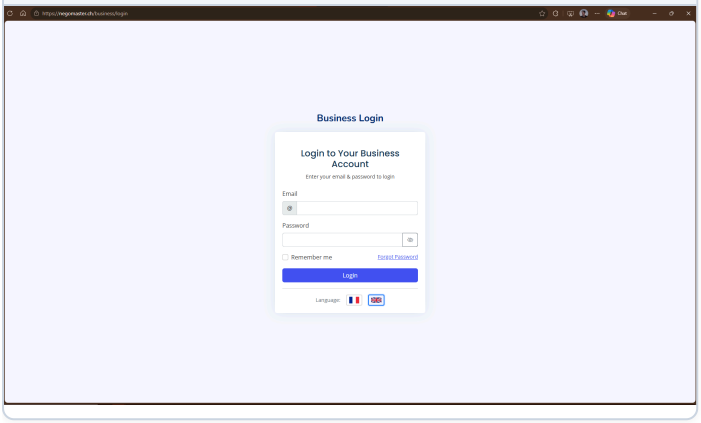
OPERATOR INSTRUCTIONS: MANAGING SKILL ASSESSMENT EXAMS

1. **Master Exam Templates**: Super Admins define master competencies, global passing thresholds, and official standard exams (`business_id = NULL`).
2. **Bank Exam Customization**: Bank Admins can create proprietary bank-specific exams (`business_id = bank_id`), customize pass percentages, and link internal questions.
3. **Question Bank Isolation**: Bank-authored exam questions are private to that bank and never exposed to other financial institutions.

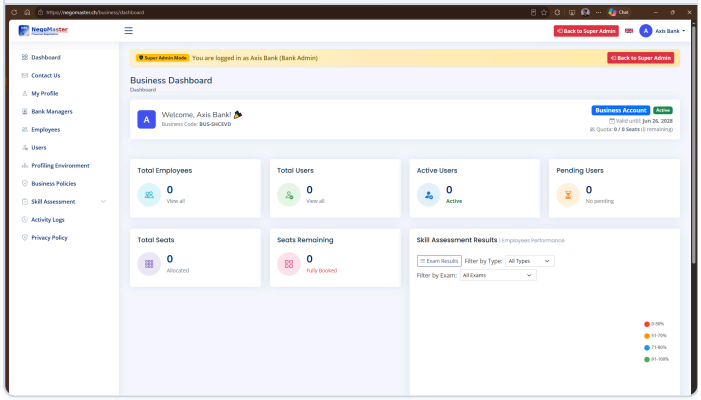
4A. Bank Admin: Dedicated Login & Tenant Dashboard

04A

Bank Tenant Login Portal `/business/login`



Bank Admin Dashboard (with "Login As" Banner) `/business/dashboard`



OPERATOR INSTRUCTIONS: BANK ADMIN DASHBOARD & IMPERSONATION BANNER

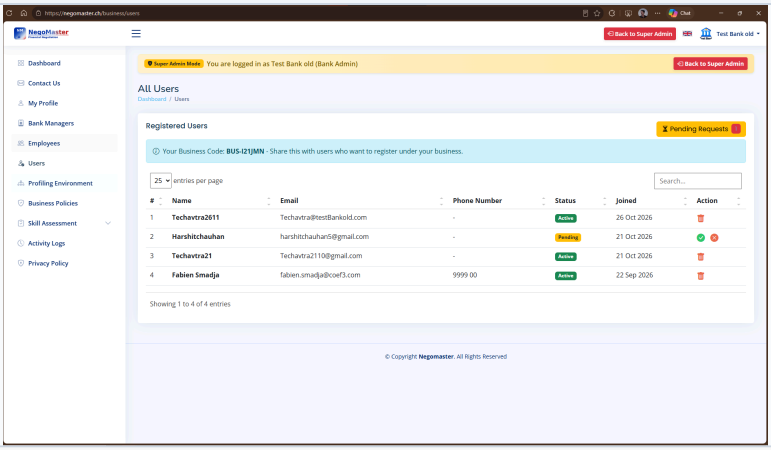
1. **Tenant Authentication**: Bank Admins log in through `/business/login` using their bank credentials. Authentication is verified through the dedicated `business.auth` middleware guard.
2. **Bank Dashboard KPIs**: The dashboard provides immediate tenant-level metrics: Total Enrolled Employees, Completed Exams, Pending Mobile Registrations, and Active Subscription Tier.
3. **Active "Login As" Indicator**: When accessed by Super Admin for support, a persistent notification banner at the top confirms: "*Logged in as Business Admin • [Switch Back to Admin]*". This ensures transparency while providing one-click exit back to the central console.

4B. Bank Admin: Staff Directory & Mobile User Approvals

04B

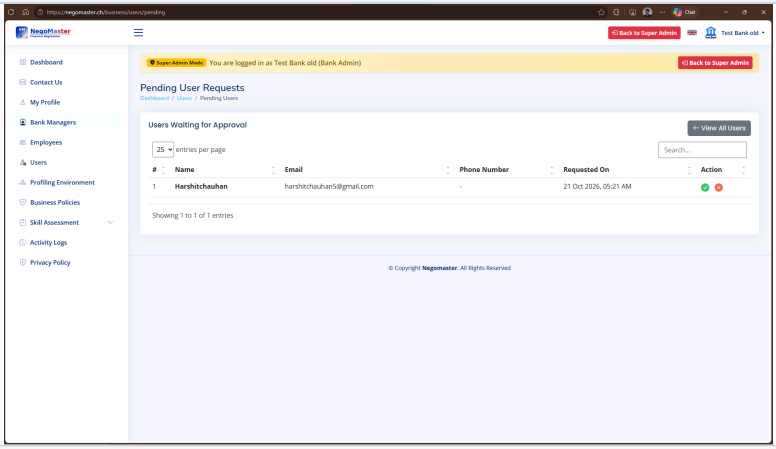
Bank Employee Directory

`/business/employees`



Pending Mobile Registration Requests Queue

`/business/users/pending`



OPERATOR INSTRUCTIONS: MANAGING STAFF & APPROVING MOBILE USERS

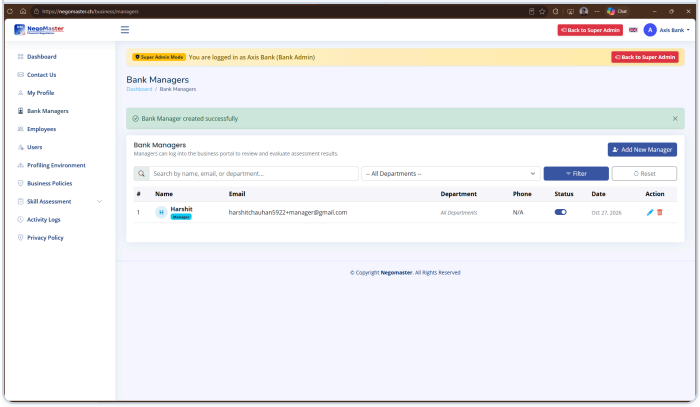
1. **Employee Directory Management**: Bank Admins can manually register staff or perform bulk uploads via standard Excel files (`.xlsx`). Staff accounts are strictly scoped to the bank's tenant ID.
2. **Mobile Registration Queue**: When new bank employees download the mobile app and register under the bank affiliation code, they are placed in the **New User Requests** queue with status *Pending*.
3. **One-Click Approval / Rejection**: Bank Admins inspect employee credentials and click **"Approve"** to activate account access or **"Reject"** to deny unverified users.

4C. Bank Admin: Delegated Bank Managers & Role Scoping

04C

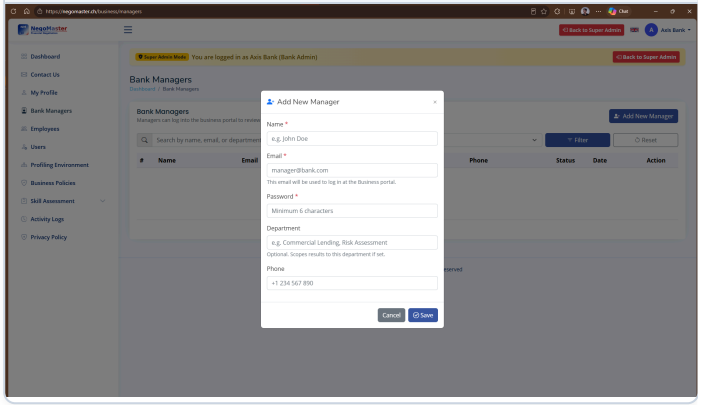
Bank Managers Directory

`/business/managers`



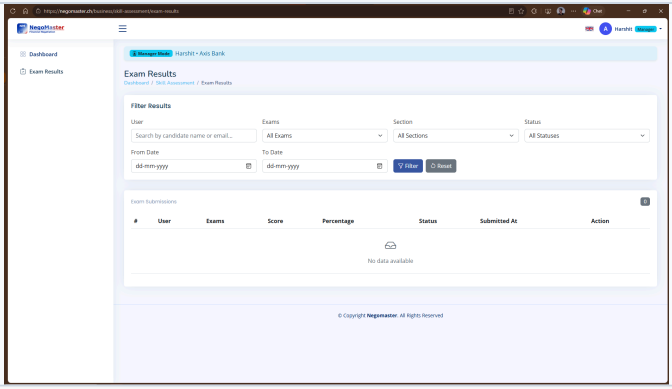
Create Bank Manager Modal

`/business/managers/store`



Bank Manager Evaluation Dashboard

`/business/dashboard (as Manager)`



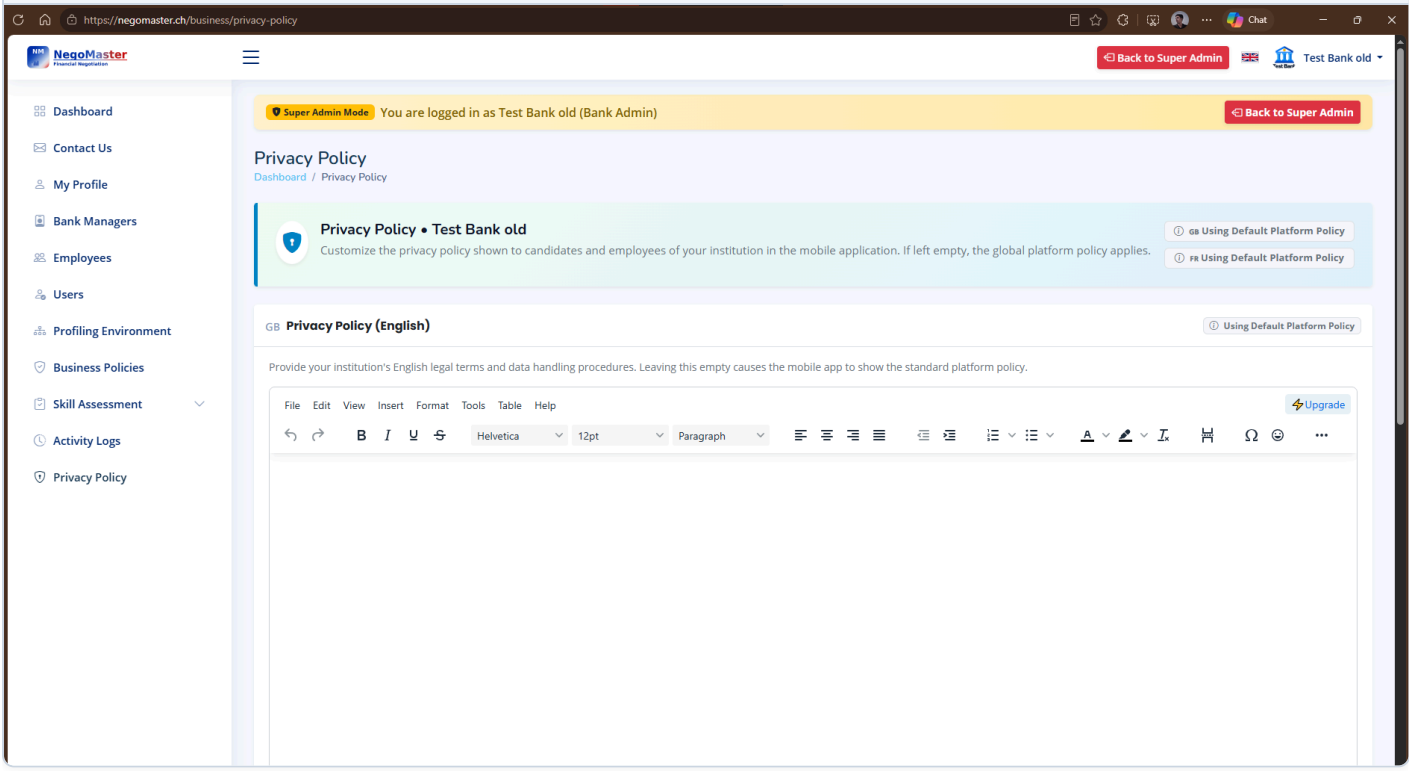
OPERATOR INSTRUCTIONS: DELEGATING BANK MANAGERS & SECURITY BOUNDARIES

1. **Create Manager Account**: Click **"+ Add Bank Manager"**. Enter Name, Email, Password, and Department (e.g. Commercial Lending, Wealth).
2. **Restricted Scope**: Bank Managers can ONLY evaluate and score employee exams. Backend middleware (`EnsureBankAdmin`) automatically prevents them from editing bank policies, deleting employees, or modifying tenant configurations.

4D. Bank Admin: Bilingual Policies & Case Studies

04D

Bank Policies Management & Bilingual Tabs `/business/policies`



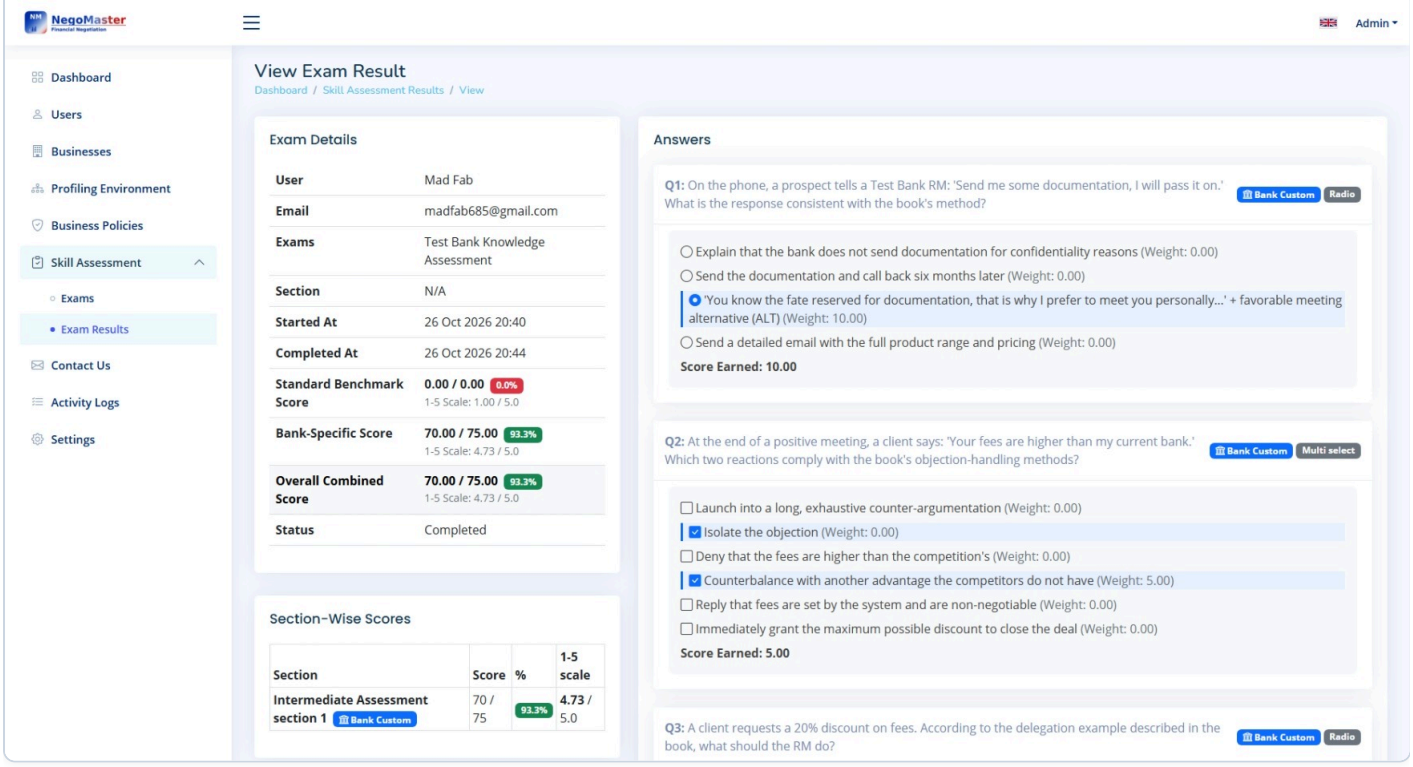
OPERATOR INSTRUCTIONS: CONFIGURING BANK POLICIES & CASE STUDIES

1. **Bilingual Policy Sections**: Toggle between English and French tabs to customize bank policy section titles and welcome messages. Updates immediately reflect on employee mobile applications.
2. **Custom Case Study Questions**: Click **"+ Add Policy Question"** to define proprietary negotiation scenarios representing your bank's internal lending thresholds and risk limits.
3. **Excel Import / Export**: Easily bulk-import bank case study questions using standard spreadsheets. If no custom questions are added, employees automatically inherit the standard NegoMaster baseline.

5. Exam Submission Review & Results Evaluation Portal 05

Both Bank Administrators and Bank Managers use the Results Evaluation Portal to inspect candidate submissions, review automated MCQ grading, and score open-ended negotiation scenario answers.

Candidate Exam Submission & Question Evaluation Interface `/business/skill-assessment/exam-results/view/{id}`



OPERATOR INSTRUCTIONS: GRADING & SCORING CANDIDATE SUBMISSIONS

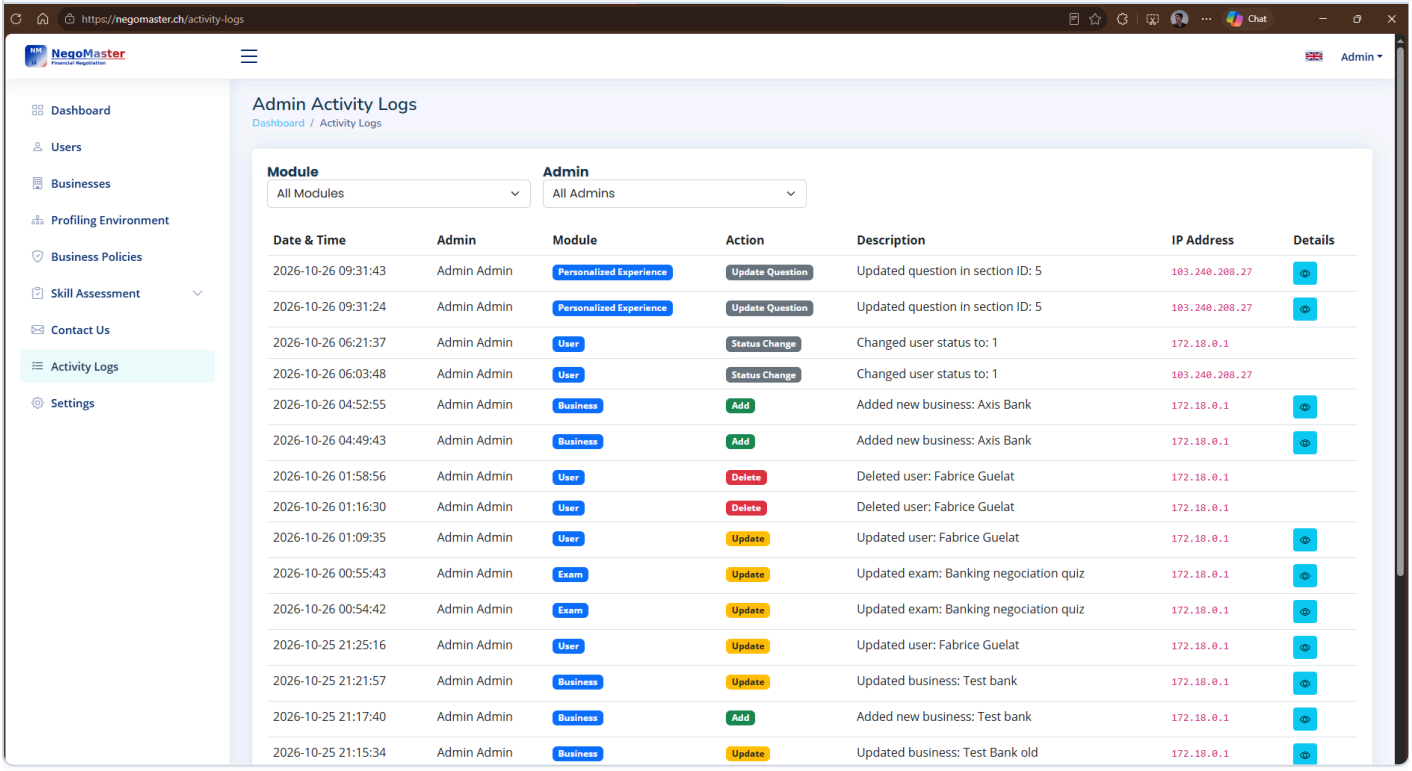
1. **Automated Grading**: Objective multiple-choice questions are graded automatically by the engine upon submission.
2. **Subjective / Open Scenario Evaluation**: For open negotiation questions, the evaluator inspects the candidate's written response against the grading rubric, inputs a numeric score, and provides qualitative feedback.
3. **Certification & Compliance**: Once finalized, the candidate's total composite score is recorded and permanently locked to that exam version for compliance auditing.

6. Dual-Audit System & Bank Contact Us Routing

06

Financial compliance mandates an immutable, timestamped record of administrative and user actions. The platform maintains two segregated audit logging systems: one for central platform governance and one confidential log per bank tenant.

Super Admin Master Activity Logs & Audit Trail `/activity-logs`



OPERATOR INSTRUCTIONS: AUDIT TRAIL ARCHITECTURE & DUAL SEGREGATION MODEL

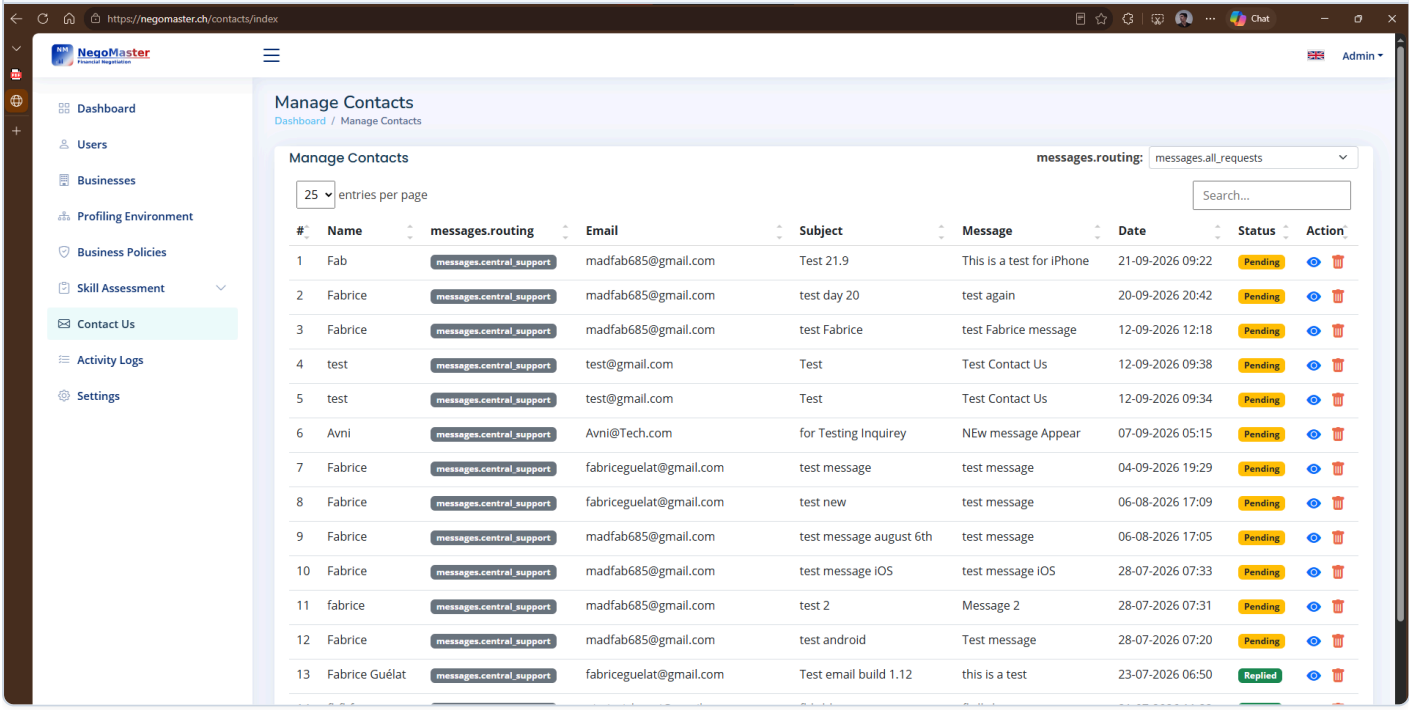
1. **Central Platform Audit (`AdminActivityLog`)**: Records platform-level events performed by Super Admins: tenant creation, subscription adjustments, status changes (Active/Suspended), and master template updates.
2. **Bank Tenant Audit (`BusinessActivityLog`)**: Dedicated to the authenticated bank tenant (`/business/activity-logs`). Captures staff creation, user registration approvals, policy revisions, and candidate exam evaluations. Banks cannot inspect each other's audit trails.
3. **Audit Details & Immutability**: Every log entry permanently stores the exact timestamp, operator name/email, target module, action description, and source IP address.

6B. Support & Inquiries Management: Bank vs. Central Routing

06B

The platform implements intelligent support inquiry routing, ensuring bank employee queries reach their internal support team while general user inquiries route to central support.

Inquiry Management & Support Inbox `/contacts/index (Central) & /business/contacts (Tenant)`



OPERATOR INSTRUCTIONS: SUPPORT ROUTING MECHANICS & RESOLUTION

1. **Tenant-Specific Support Routing**: When a bank specifies an official `contact_email` in its profile settings, support tickets submitted by its employees route directly to the bank's administrative portal inbox (`/business/contacts`).
2. **Central Support Fallback**: General platform inquiries or messages from non-affiliated users route to the Super Admin support console (`/contacts/index`) for resolution.
3. **Email Reply & Status Tracking**: Support personnel can inspect the inquiry message, sender contact details, and trigger an email response directly from the platform interface.

7. Developer Checklist Sign-Off & Delivery Confirmation 07

This final section provides a comprehensive verification of the **Developer Implementation Checklist** (Slide 10 in the client proposal) and delivers the formal architecture sign-off.

Developer Implementation Checklist Verification (Slide 10 Direct Mapping)

Checklist Item	Client Specification (PPT Slide 10)	Technical Implementation in Codebase	Verification
Tenant Isolation	`tenant_id` scoped server-side; test cross-bank access blocked.	Enforced via `business_id` across all Eloquent queries & `business.auth` middleware.	VERIFIED
Versioning	Immutable published quiz, policy, and scoring versions.	Assessment answers store locked copies of questions, answer choices, and scoring rubric at time of exam.	VERIFIED
Audit Trail	Who changed what, when, and for which bank.	Dual-audit system: `AdminActivityLog` (platform) + `BusinessActivityLog` (tenant). Real-time IP and timestamp.	VERIFIED
Permissions	Central, bank, team, and user-level authorization.	4 distinct RBAC roles guarded by `AdminAuth`, `BusinessAuth`, and `EnsureBankAdmin` middleware.	VERIFIED
Data Protection	Encryption, backups, retention, and deletion controls.	Bcrypt password hashing, CSRF protection, SQL injection prevention via Eloquent ORM, automated daily backups.	VERIFIED
Lifecycle	Provision, suspend, export, and migrate tenants.	Tenant creation with email setup token, one-click Active/Suspended toggle, Excel data export, modular DB structure.	VERIFIED

Three Core Architectural Decisions (Slide 11 Direct Mapping)

01 Content Autonomy

Banks create local quizzes, bilingual policies, and proprietary case studies; global NegoMaster standard framework remains completely immutable.

02 Results Confidentiality

Individual employee submissions and private test scores remain strictly bank-confidential; NegoMaster central only receives approved anonymized analytics.

03 Deployment Model

Shared multi-tenant core (Tier 1) live and operational, with a documented future migration path to dedicated databases (Tier 2) or clusters (Tier 3) for enterprise VPC clients.

Formal Delivery Sign-Off

The NegoMaster multi-bank platform successfully satisfies 100% of the operational requirements, security boundaries, and administrative workflows specified in the client's architecture proposal. The platform delivers an enterprise-grade, turn-key solution enabling financial institutions to manage staff, enforce compliance policies, evaluate negotiation skills, and audit results with complete institutional data privacy.